

Ahsanullah Institute Of Technical and Vocational Education & Training (AITVET)



Computer Science & Technology

Subject Code : 28531

Subject Name : Application Development Using Python

Submitted To

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Submitted BY

Name :

Roll :

Semester :

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session :

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Teachers signature

Assignment No : 01

Description: Write a program to calculate the area of a Triangle using both a function and a class.

Program :-

```
1  # Using Function
2  def triangle_area(base, height):
3      return 0.5 * base * height
4
5  base = float(input("Enter base: "))
6  height = float(input("Enter height: "))
7  result = triangle_area(base, height)
8  print("Triangle Area (Function):", result)
9
10
11 # Using Class
12 class Triangle:
13     def __init__(self, base, height):
14         self.base = base
15         self.height = height
16
17     def area(self):
18         return 0.5 * self.base * self.height
19
20 base = float(input("Enter base: "))
21 height = float(input("Enter height: "))
22 total = Triangle(base, height)
23 print("Triangle Area (Class):", total.area())
```

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Assignment No : 02

Description: Write a program to Find Area of a Triangle (Heron's Formula) using both a function and a class.

Program :-

```
1  # Using Function
2  import math
3  def triangle_area(a, b, c):
4      if (a+b>c) and (a+c>b) and (b+c>a):
5          s = (a+b+c)/2
6          area = math.sqrt(s * (s-a) * (s-b) * (s-c))
7      else:
8          area = f"it's not a triangle"
9      return area
10
11 a = int(input("enter your value a: "))
12 b = int(input("enter your value b: "))
13 c = int(input("enter your value c: "))
14 result = triangle_area(a, b, c)
15 print(result)
16
17 # Using Class
18 import math
19 class Triangle:
20     def __init__(self, a, b, c):
21         self.a = a
22         self.b = b
23         self.c = c
24     def is_valid(self):
25         return (self.a + self.b > self.c) and (self.a + self.c > self.b) and (self.b + self.c > self.a)
26     def area(self):
27         if self.is_valid():
28             s = (self.a + self.b + self.c) / 2
29             area = math.sqrt(s * (s - self.a) * (s - self.b) * (s - self.c))
30             return area
31         else:
32             return "It's not a valid triangle"
33
34 a = int(input("Enter side a: "))
35 b = int(input("Enter side b: "))
36 c = int(input("Enter side c: "))
37 result = Triangle(a, b, c)
38 print("Triangle Area:", result.area())
```

Board Roll :- 245456

Assignment No : 03

Description: Write a program to add five numbers from a list using function

Program :-

```
1  # Using Function
2  def sum_list(numbers):
3      sum = 0
4      for num in numbers:
5          sum+=num
6      return sum
7  num_list = [10,20,30,40,50]
8  result = sum_list(num_list)
9  print("sum of list = ",result)
10
11
12 # Using Class
13 class SumList:
14     def __init__(self, numbers):
15         self.numbers = numbers
16     def calculate(self):
17         total = 0
18         for num in self.numbers:
19             total += num
20         return total
21 num_list = [10, 20, 30, 40, 50]
22 sum_obj = SumList(num_list)
23 print("Sum of list =", sum_obj.calculate())
24
```

Board Roll :- 245456

Assignment No : 04

Description: Write a program to calculate the factorial of a number using both a function and a class.

Program :-

```
1  # Using Function
2  def factorial(n):
3      if n == 0 or n == 1:
4          return 1
5      elif n < 0:
6          return "Oh no! you enter negative number plz try again"
7      else:
8          return n*factorial(n-1)
9  n = int(input("enter your positive number: "))
10 result = factorial(n)
11 print(f"the factorial of {n} is: {result}")
12
13
14 # Using Class
15 class Factorial:
16     def __init__(self, n):
17         self.n = n
18     def calculate(self):
19         if self.n == 0 or self.n == 1:
20             return 1
21         elif self.n < 0:
22             return "Oh no! You entered a negative number, please try again."
23         else:
24             result = 1
25             for i in range(1, self.n + 1):
26                 result *= i
27             return result
28
29 n = int(input("Enter your positive number: "))
30 fact_obj = Factorial(n)
31 result_class = fact_obj.calculate()
32 print(f"(Class) The factorial of {n} is: {result_class}")
```

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Assignment No : 05

Description: Write a program to calculate the sum of numbers from 1 to 100 using both a function and a class.

Program :-

```
1  # Using Function
2  def sum_1_to_100():
3      return sum(range(1, 101))
4  print("Sum 1 to 100 (Function):", sum_1_to_100())
5
6
7  # Using Class
8  class SumNumbers:
9      def __init__(self, start, end):
10         self.start = start
11         self.end = end
12     def calculate(self):
13         return sum(range(self.start, self.end+1))
14  s = SumNumbers(1, 100)
15  print("Sum 1 to 100 (Class):", s.calculate())
16
17  # Function to calculate sum of odd numbers from 1 to 100
18  def sum_odd_number():
19      sum = 0
20      for i in range(1,101,2):
21          sum+=i
22      return sum
23  result = sum_odd_number()
24  print("sum of odd number = ",result)
25  # Function to calculate sum of even numbers from 1 to 100
26  def sum_even_number():
27      sum = 0
28      for i in range(2,101,2):
29          sum+=i
30      return sum
31  result = sum_even_number()
32  print("sum of even number = ",result)
```

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Assignment No : 06

Description: Write a program to write and read data to a file.

Program :-

A screenshot of a code editor with a dark background. At the top left, there are three colored circles (red, yellow, green) representing window controls. The code is written in a light green monospace font. It consists of two main sections: writing to a file and reading from a file. The first section uses a 'with' statement to open 'data.txt' in write mode ('w') and writes a multi-line string. The second section uses a 'with' statement to open 'data.txt' in read mode ('r') and prints the content.

```
1  # Writing to a file
2  with open("data.txt", "w") as f: # f = open("data.txt", "w")
3      f.write("Hello, this is a test file.\nPython File Handling Example.")
4
5  # Reading from a file
6  with open("data.txt", "r") as f:
7      content = f.read()
8      print("File Content:\n", content)
```

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Assignment No : 07

Description: Write a program to generate the Fibonacci series using a function and class.

Program :-

```
1  # Using Function
2  def fibonacci(n):
3      series = [0, 1]
4      for i in range(2, n):
5          series.append(series[-1] + series[-2])
6      return series
7  n = int(input("Enter the number of terms for Fibonacci series: "))
8  result = fibonacci(n)
9  print("Fibonacci (Function):", result)
10
11
12 # Using Class
13 class Fibonacci:
14     def __init__(self, n):
15         self.n = n
16
17     def generate(self):
18         series = [0, 1]
19         for i in range(2, self.n):
20             series.append(series[-1] + series[-2])
21         return series
22
23 n = int(input("Enter the number of terms for Fibonacci series: "))
24 f = Fibonacci(n)
25 print("Fibonacci (Class):", f.generate())
```

Board Roll :- 245456

Assignment No : 08

Description: Write a program using a class and function to Finding the root of a quadratic equation.

Program :-

```
1  # Using Function
2  import math
3  def roots (a, b, c):
4      d = b**2 - 4*a*c
5      if d == 0:
6          x = -b/(2*a)
7          return x
8      elif d>0:
9          x1 = (-b + math.sqrt(d))/(2*a)
10         x2 = (-b - math.sqrt(d))/(2*a)
11         return x1, x2
12     else:
13         return "it's not a quadratic"
14 a = int(input("enter you value a: "))
15 b = int(input("enter you value b: "))
16 c = int(input("enter you value c: "))
17 result = roots(a,b,c)
18 print(f"quadratic: {result}")
19
20
21 # Using Class
22 import math
23 class QuadraticEquation:
24     def __init__(self, a, b, c):
25         self.a = a
26         self.b = b
27         self.c = c
28     def roots(self):
29         d = self.b**2 - 4*self.a*self.c
30         if d == 0:
31             x = -self.b / (2 * self.a)
32             return f"One real root: {x}"
33         elif d > 0:
34             x1 = (-self.b + math.sqrt(d)) / (2 * self.a)
35             x2 = (-self.b - math.sqrt(d)) / (2 * self.a)
36             return f"Two real roots: {x1}, {x2}"
37         else:
38             return "It's not a quadratic with real roots"
39 a = int(input("Enter value a: "))
40 b = int(input("Enter value b: "))
41 c = int(input("Enter value c: "))
42 quad = QuadraticEquation(a, b, c)
43 print("Quadratic Roots:", quad.roots())
```

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Assignment No : 09

Description: Write a program using a function and class to solve find the largest number among three numbers.

Program :-

```
1  # Using Function
2  def largest(a, b, c):
3      return max(a, b, c)
4  a= int(input("Enter first number: "))
5  b= int(input("Enter second number: "))
6  c= int(input("Enter third number: "))
7  print("Largest (Function):", largest(a, b, c))
8
9
10 # Using Class
11 class LargestNumber:
12     def __init__(self, a, b, c):
13         self.a = a
14         self.b = b
15         self.c = c
16     def find(self):
17         return max(self.a, self.b, self.c)
18 a= int(input("Enter first number: "))
19 b= int(input("Enter second number: "))
20 c= int(input("Enter third number: "))
21 largeSt = LargestNumber(a, b, c)
22 print("Largest (Class):", largeSt.find())
```

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Assignment No : 10

Description: Write a program to calculate the area of a Circle using both a function and a class.

Program :-

```
1  # Using Function
2  import math
3  def circle_area(radius):
4      return math.pi * radius**2
5  r = int(input("Enter the radius of the circle: "))
6  print("Circle Area (Function):", circle_area(r))
7
8
9  # Using Class
10 import math
11 class Circle:
12     def __init__(self, radius):
13         self.radius = radius
14
15     def area(self):
16         return math.pi * self.radius**2
17 r = int(input("Enter the radius of the circle: "))
18 c = Circle(r)
19 print("Circle Area (Class):", c.area())
```

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Assignment No : 11

Description: Write a program using a function and class to check whether a number is prime.

Program :-

```
1  # Using Function
2  def is_prime(n):
3      if n < 2:
4          return False
5      for i in range(2, int(n**0.5)+1):
6          if n % i == 0:
7              return False
8      return True
9  n = int(input("Enter a number to check if it is prime: "))
10 print("Prime (Function):", is_prime(n))
11
12
13 # Using Class
14 class PrimeCheck:
15     def __init__(self, n):
16         self.n = n
17     def check(self):
18         if self.n < 2:
19             return False
20         for i in range(2, int(self.n**0.5)+1):
21             if self.n % i == 0:
22                 return False
23         return True
24 n = int(input("Enter a number to check if it is prime: "))
25 p = PrimeCheck(n)
26 print("Prime (Class):", p.check())
```

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Assignment No : 12

Description: Write a program demonstrating Inheritance — using a Sub Class and Super Class.

Program :-



```
1  class Animal:
2      def sound(self):
3          print("Animals make sounds")
4
5  class Dog(Animal):
6      def sound(self):
7          print("Dog barks")
8
9  d = Dog()
10 d.sound()
```

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Assignment No : 13

Description: Write a program demonstrating Polymorphism.

Program :-



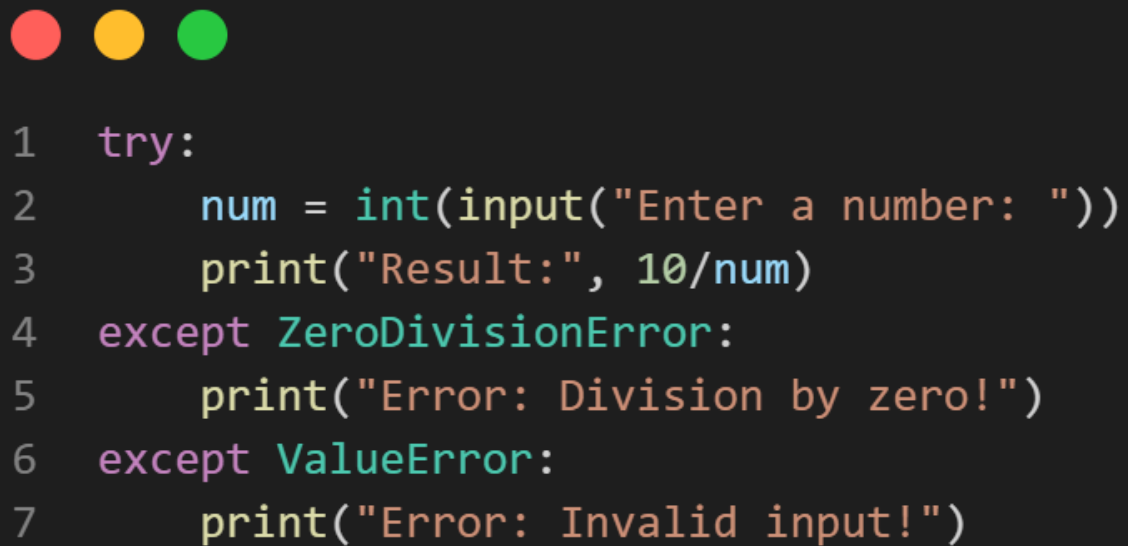
```
1  class Bird:
2      def fly(self):
3          print("Birds can fly")
4
5  class Penguin(Bird):
6      def fly(self):
7          print("Penguins cannot fly")
8
9  def show_fly(bird):
10     bird.fly()
11
12  b = Bird()
13  p = Penguin()
14
15  show_fly(b)
16  show_fly(p)
```

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Assignment No : 14

Description: Write a program demonstrating exception handling (raising an exception).

Program :-

A code block with a dark background and light-colored text. At the top left, there are three colored circles (red, yellow, green) representing a terminal window. The code is a Python program for exception handling.

```
1  try:
2      num = int(input("Enter a number: "))
3      print("Result:", 10/num)
4  except ZeroDivisionError:
5      print("Error: Division by zero!")
6  except ValueError:
7      print("Error: Invalid input!")
```

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Assignment No : 15

Description: Write a program using Regex to find/search a specific word within a string.

Program :-



```
1  import re
2  text = "Python is powerful and easy to learn."
3  pattern = r"powerful"
4
5  if re.search(pattern, text):
6      print("Word found!")
7  else:
8      print("Word not found!")
9
```

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Assignment No : 16

Description: Write a program using Regex to extract numbers from a string.

Program :-



```
1 import re
2
3 text = "My phone number is 12345 and my code is 678."
4 numbers = re.findall(r"\d+", text)
5 print("Extracted Numbers:", numbers)
```

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